

Motivation

Broader context:

CO₂ has silently crossed **420 ppm**. The world is heating, oceans are acidifying, and climate tipping points are accelerating. Yet most direct air capture and CO₂-to-fuel systems remain **fragile, costly, and disconnected from reality**. They fail under **ambient conditions**, require **high energy input**, and **break down** when scaled.

Game changing breakthrough:

Porous organic polymers are **solid, stable, and tuneable**. They can trap CO₂ from thin air and convert it into formic acid with **high efficiency and unmatched reliability**.

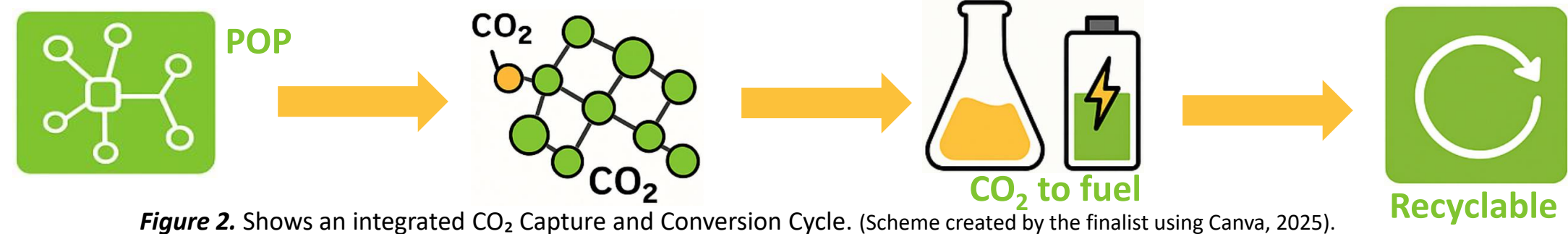


Figure 2. Shows an integrated CO₂ Capture and Conversion Cycle. (Scheme created by the finalist using Canva, 2025).

Current challenges:

- Commercial DAC units capture less than 0.01 percent of global emissions.
- Solvent-based systems are corrosive, expensive, and unstable.
- Catalysts degrade or lose selectivity under real operating conditions.
- Hydrogen evolution reaction competes, blocking CO₂ reduction.
- No existing platform unites capture, conversion, and storage in a scalable design.

Research Background

- In 2023, global DAC facilities captured less than 10,000 tons of CO₂, far below climate goals (Mongabay, 2024). Climeworks Orca plant in Iceland used DAC to remove about 4,000 tons annually but faced high energy demands and costs of \$750 to \$1,000 per ton. While the LEILAC project captured CO₂ from cement industries through indirect calcination, yet scalability remains limited by the high thermal requirement near 900°C (LEILAC, 2023). These challenges highlight the urgent need for modular, scalable, and low-cost capture technologies.

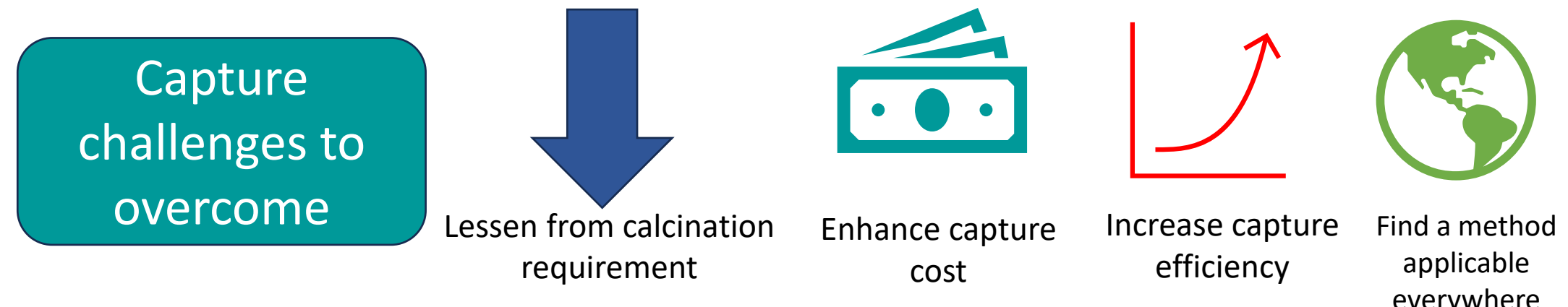


Figure 2. schematic representation of the challenges in the capture process. (Scheme created by the finalist using powerpoint, 2025).

- Several electrochemical methods have reported poor efficiencies for CO₂-to-formic acid conversion. For example, a study using SnO₂ catalysts achieved only 26.7% Faradaic efficiency at -1.3 V with low selectivity and significant hydrogen evolution (Zhou et al., *Electrochimica Acta*, 2022). Similarly, an In-based system operated at -1.0 V yielded just 18.4% efficiency with unstable current densities and rapid catalyst degradation (Lee et al., *J. CO₂ Util.*, 2021). These results reflect ongoing struggles with energy loss and product instability in current systems.



Figure 3. schematic diagram of the challenges in the conversion process. (Scheme created by the finalist using powerpoint, 2025).

Objectives

- Use amine-modified polymers to enhance direct CO₂ capture from air, addressing limitations in selectivity, surface area, and regeneration seen in conventional materials.
- Overcome the low efficiency and instability of CO₂ conversion by using metal-porphyrin porous polymers with tuneable structure, improved Faradaic efficiency, reduced HER, and stable performance.
- Offering **the first ever one of a kind** integrated direct air capture and electrochemical conversion system using novel, cost-effective, efficient, and highly scalable POPs.

Hypothesis

If two tailored porous organic polymers are used separately for CO₂ capture and electrochemical conversion, then each will surpass benchmark materials in efficiency, offering a viable path toward scalable carbon utilization.

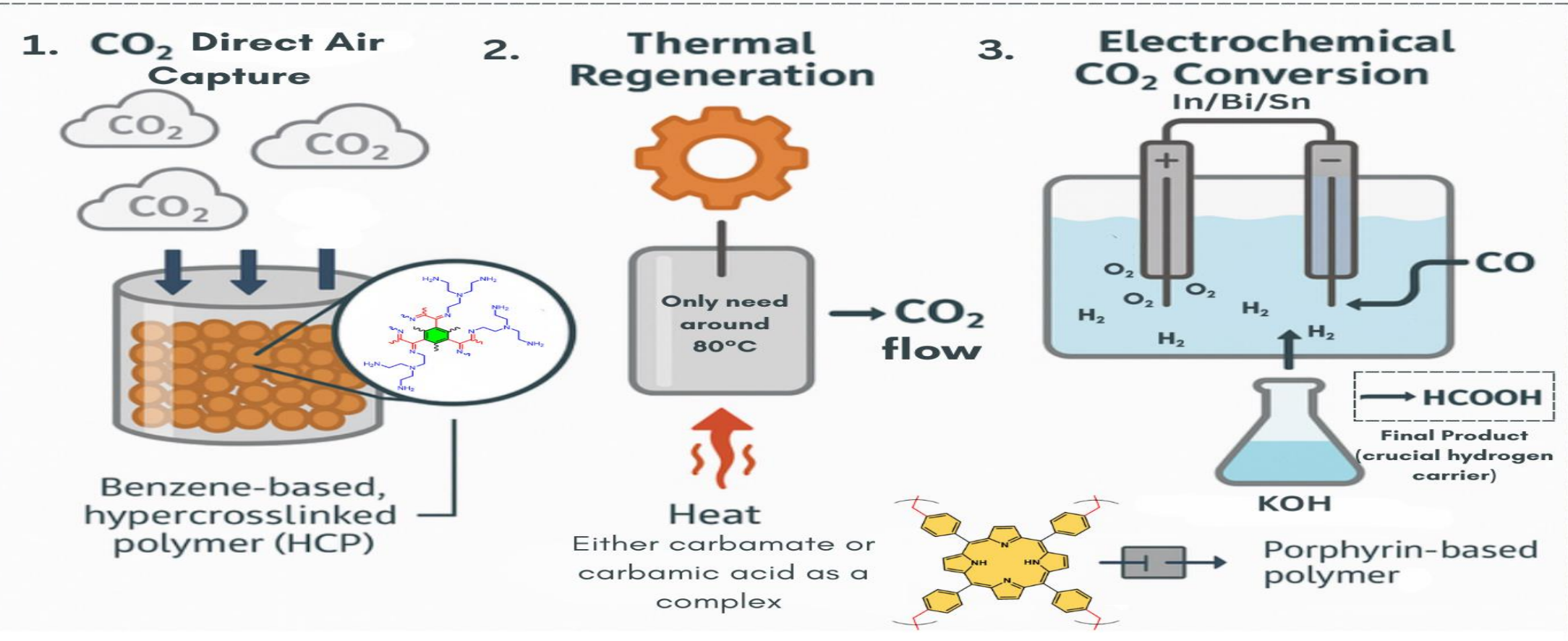


Figure 4. schematic representation of the integrated direct air capture and conversion system. (Scheme created by the finalist using canva and biorender, 2025)

Variables

Direct air capture

- Independent:** Amine loading and polymer surface area
- Dependent:** CO₂ uptake capacity (cm³/g), adsorption duration, and stability
- Controlled:** Temperature, airflow rate, and gas concentration

Conversion

- Independent:** Applied voltage and metal catalyst used
- Dependent:** Faradaic Efficiency (%) and catalyst stability
- Controlled:** Electrolyte type and cell type

Experimental Work

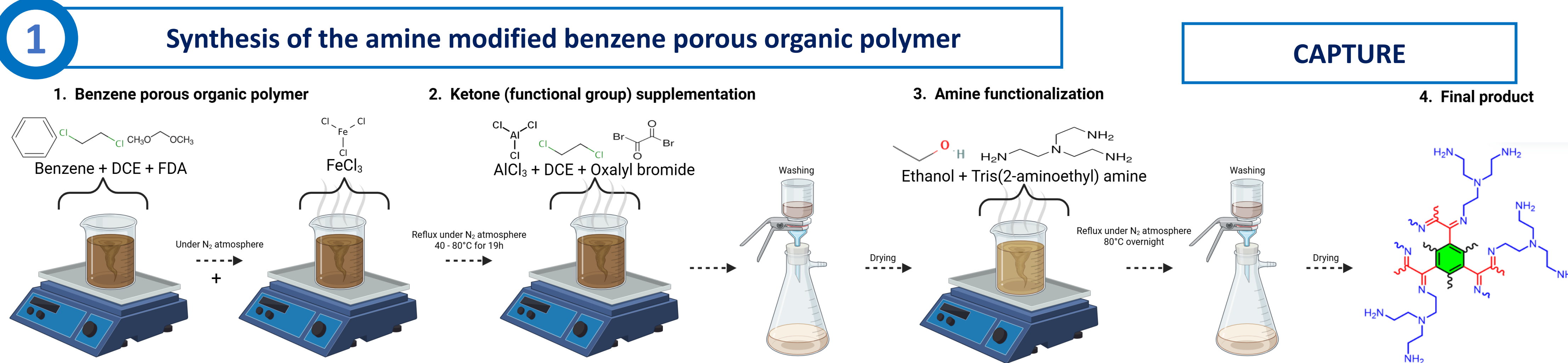


Figure 5. Schematic representation of the synthesis of the amine modified benzene porous organic polymer. Figure 5 was created by the finalist using biorender, 2025.

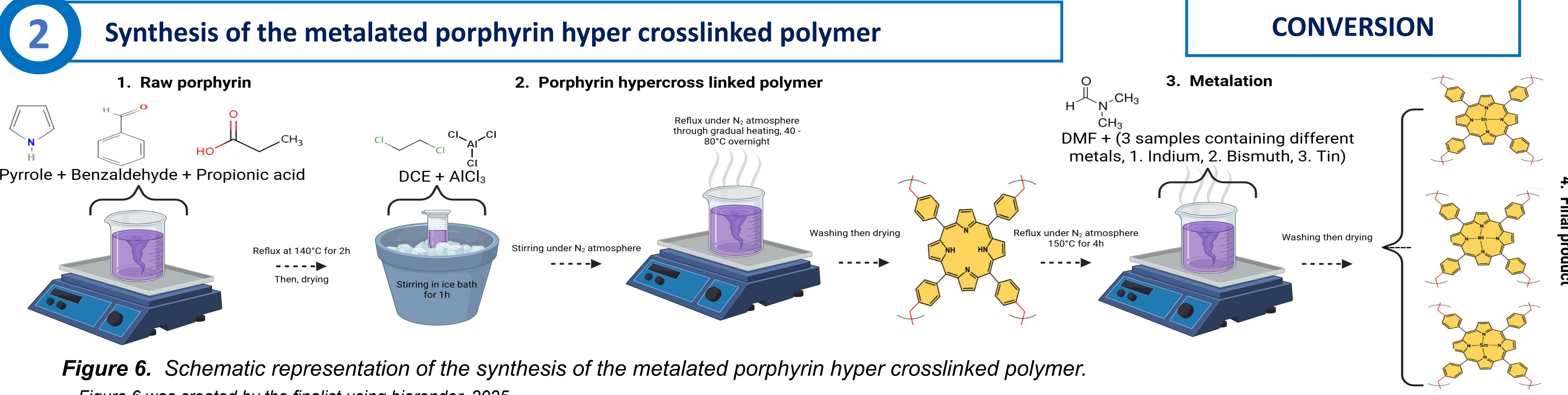
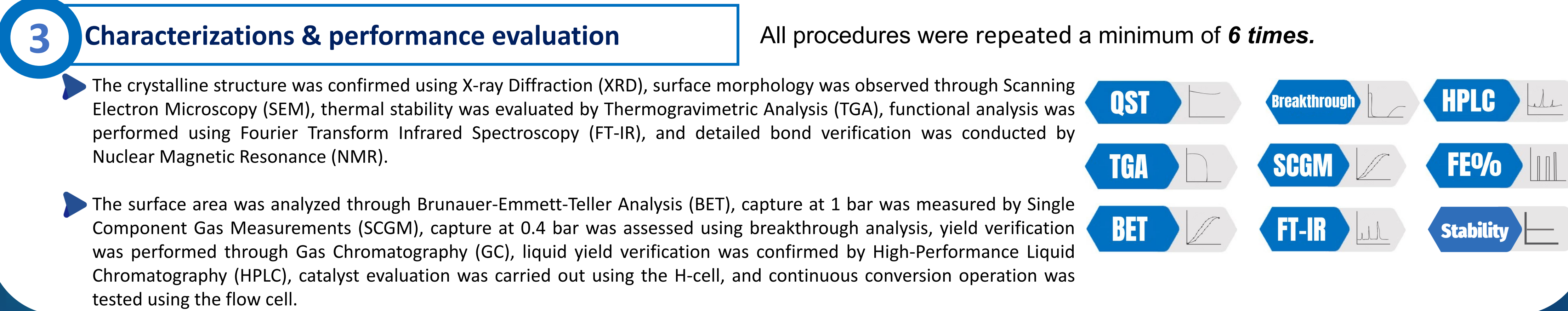


Figure 6. Schematic representation of the synthesis of the metalated porphyrin hyper crosslinked polymer. Figure 6 was created by the finalist using biorender, 2025.



Data and Graphs

Characterization

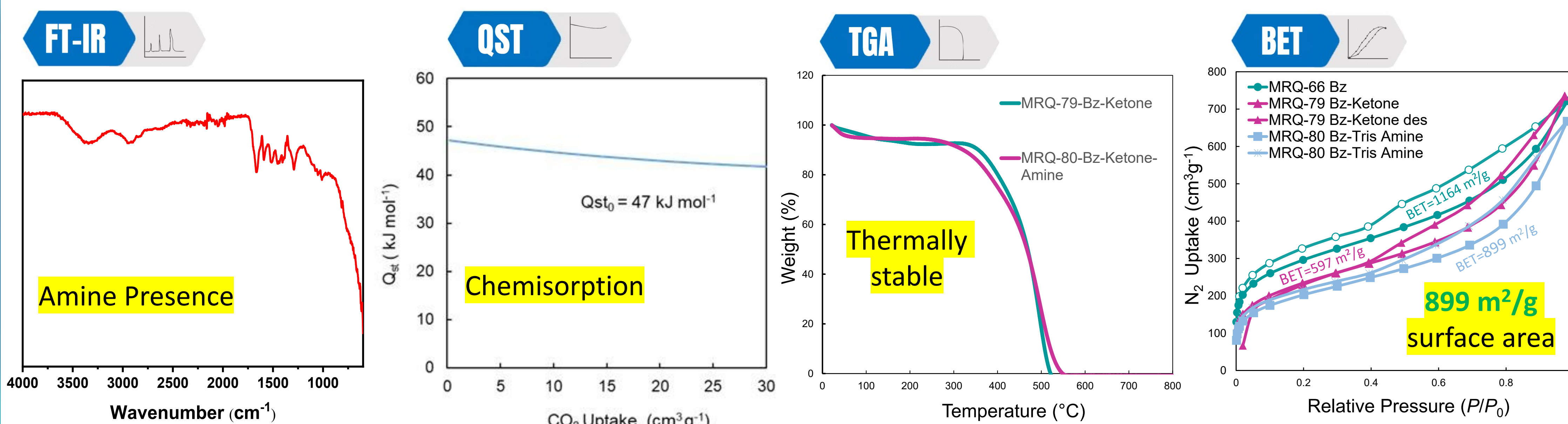


Figure 7. Presents the FT-IR graph proving the amine presence. (Graph created by the finalist using excel, 2025).

Figure 8. QST illustrates a chemisorption interaction which is perfect for DAC. (Graph created by the finalist using excel, 2025).

Figure 9. Proves the thermal stability of the material. (Graph created by the finalist using excel, 2025).

Figure 10. Presents surface area of the benzene POP, before & after modification. (Graph created by the finalist using excel, 2025).

Performance Analysis

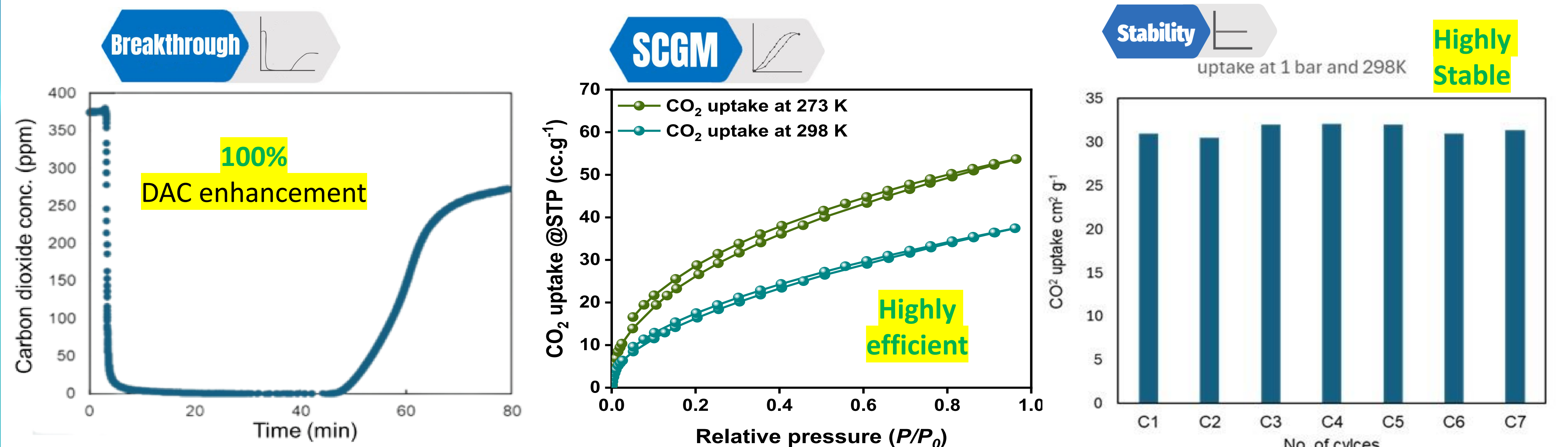


Figure 11. The Bz-POP was able to sustain capacity for ~ 40 min while also achieving high capture capacity. (Graph created by the finalist using excel, 2025).

Figure 12. Bz-POP was able to achieve over 53 cm³/g at 273 K and over 38 cm³/g at 298 K. (Graph obtained by the finalist using SCGM instrument, 2025).

Figure 13. Shows highly stable CO₂ capture under 1 bar for over 7 cycles. (Graph created by the finalist using excel, 2025).

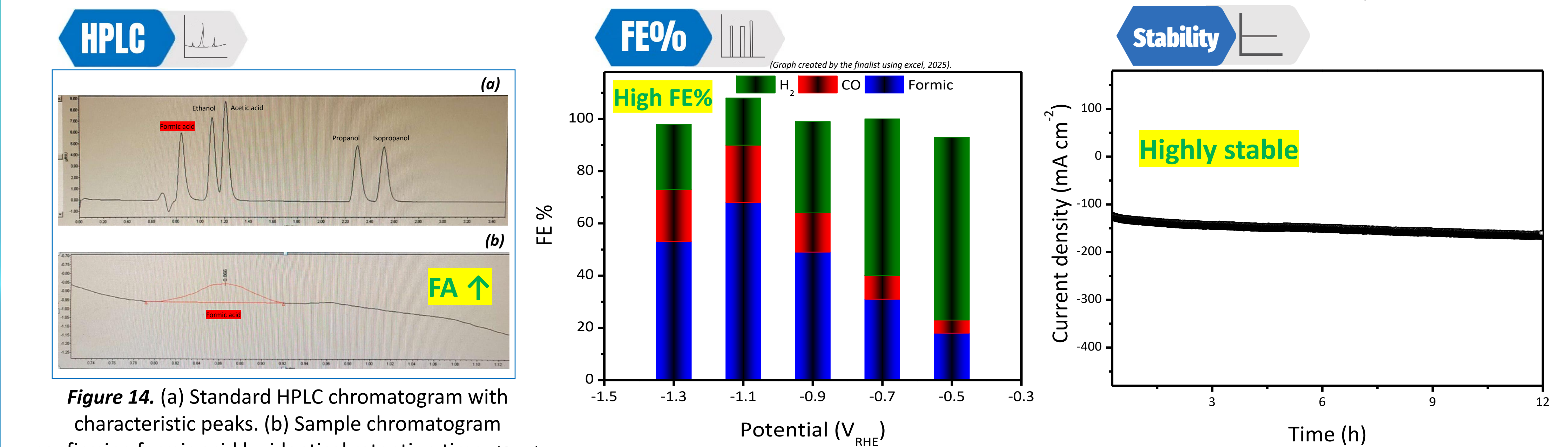


Figure 14. (a) Standard HPLC chromatogram with characteristic peaks. (b) Sample chromatogram confirming formic acid by identical retention time. (Graph obtained by the finalist using an HPLC instrument, 2025).

Figure 15. Proves that raising the current boosted formic acid production over 70%, reduced HER.

Figure 16. Presents highly stable performance for over 12 hours. (Graph created by the finalist using excel, 2025).

Results

- Captured 53.4 cm³/g of CO₂ with amine-benzene POP, achieving **100% DAC enhancement** after functionalization, and ultimately outperforming benchmark POPs by over 66%. The material showed to be **stable** for over 7 cycles. (Figure 11, 12, and 13).
- Converted CO₂ into formic acid reaching **75 FE%** with indium catalyst, increasing efficiency by **three folds** over benchmark, while maintaining over 12 h of **stable performance**. (Figure 15 and 16).
- Capture costs dropped by **70%** with amine-benzene POP, while porphyrin polymer cut conversion costs by **90%**, marking a **breakthrough** in affordable carbon solutions.
- Both the capture and conversion systems maintained over **95%** of their performance across **7 consecutive cycles and multiple setups**, proving the platform is **stable, repeatable, and scalable** for real world use.

Commercialization

As shown in Table 1, the proposed Amine-Bz-POP **outperformed benchmark material in DAC by over 66.9%**, while the metal-Porph-HCP **exceeded benchmark conversion performance by more than 294.7%**. Both POPs operating at **less than half the cost**.

Name	Function	Efficiency	Cost per gram	Reference
man-Azo-P1-POP	Capture	32 cm ³ g ⁻¹	7\$	[4]
Cu-Al Catalyst	Conversion	19%	5 - 8\$	[5]
Amine-Bz-POP (this work)	Capture	53.4 cm ³ g ⁻¹	3.61\$	-
Metal-Porph-HCP (this work)	Conversion	73%	1.1 - 1.3\$	-

(Created by finalist, 2025)

Conclusion and Application

Hypothesis proven: Tailored porous polymers captured 2x more CO₂ and produced 3x more formic acid

- The amine-functionalized polymer captured CO₂ at over twice the capacity of benchmark materials, sustaining performance for 40 minutes. The metal-porphyrin polymer converted CO₂ into formic acid with 75% Faradaic efficiency, nearly 300% better than conventional systems with maintaining lower voltage, less HER, and strong stability.
- Where past methods failed with energy loss, catalyst breakdown, and poor scalability, this platform delivers. **No solvents. No compromises.** Just solid-state real-world performance.
- Verbal agreements with industry partners are in place. Deployment begins with modular rooftop and industrial units across cities and remote zones. **This is not lab-scale. This is field-ready.** Direct air to clean fuel.

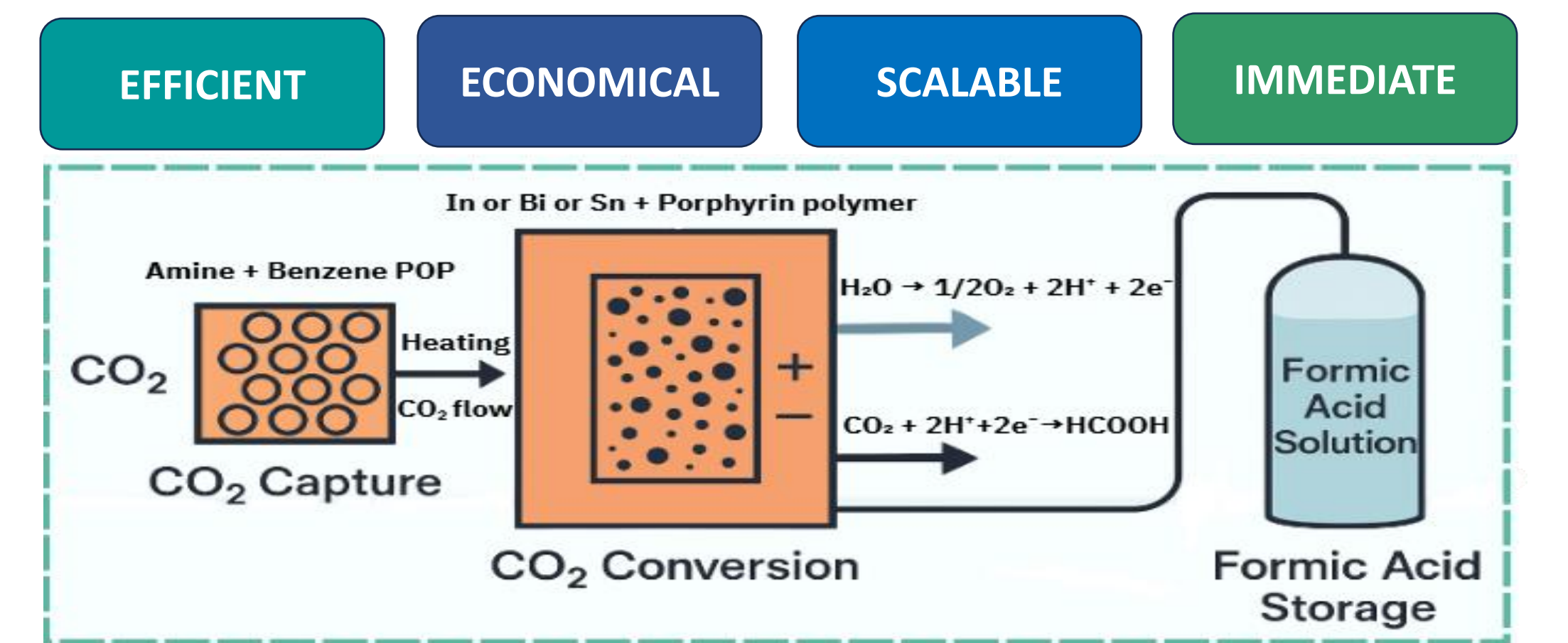


Figure 17. Shows the device/system schematic. (Scheme created by the finalist using biorender and canva, 2025).

Future Work

- Further work on the project involves:**
- Experiencing different types of amine
- Experiencing different concentrations of the catalysts
- Incorporate smart sensors for real time monitoring

References

- [1] Torres, L., Castagnino, R., Chakrabarti, S., Lima, J., Cannon, J., Kapoor, M. L., Catanoso, J., Shaw, A., Truscott, R., Tisna, R., Iverson, E., & Watch, D. (2024, December 20). *Electrochemical removal of Ocean CO2 offers potential and concerns*. Mongabay Environmental News. <https://news.mongabay.com/2024/12/electrochemical-removal-of-ocean-co2-offers-potential-and-concerns/>
- [2] *Climeworks opens world's largest plant to extract CO2 from air in Iceland | reuters*. Available at: <https://www.reuters.com/business/environment/climeworks-opens-worlds-largest-plant-extract-co2-air-iceland-2024-05-08/> (Accessed: 24 April 2025).
- [3] Hills, Thomas & Rennie, Daniel & Fennell, Paul. (2017). LEILAC: Low Cost CO₂ Capture for the Cement and Lime Industries. *Energy Procedia*. 114. 6166-6170. 10.1016/j.egypro.2017.03.1753.
- [4] Mahmoud M. Abdelnaby, Tawfik A. Saleh, Mostafa Zeama, Mahmoud Atef Abdalla, Hossam M. Ahmed, and Mohamed A. Habib.ACS Omega 2022 7 (17), 14535-14543. DOI: 10.1021/acsomega.1c05905
- [5] Nitumani Das, Ratul Paul, Duy Quang Dao, Rupak Chatterjee, Kashmiri Borah, Subhash Chandra Shit, Asim Bhaumik, and John Mondal. ACS Applied Nano Materials 2022 5 (4), 5302-5315.DOI: 10.1021/acsaanm.2c00311